Enabling Cross-Domain Semantic Data Integration with an RDF Representation of RefChem

Qingliang Li[1], Sunghwan Kim[1], Paul Thiessen[1], Jian Zhang[1], Tiejun Cheng[1], Asta Gindulyte[1], Siqian He[1], Evan Bolton[1*]

[1] National Center for Biotechnology Information, National Library of Medicine, National Institutes of Health, Bethesda, MD 20894, USA

* Corresponding author (bolton@ncbi.nlm.nih.gov)

Abstract

Chemical entity integration across heterogeneous biomedical resources remains a persistent challenge due to variability in how substances are defined, named, and contextualized. Even among curated and authoritative data sources, disagreements arise over whether two representations refer to the same chemical entity. For example, distinctions between formaldehyde as a gas and formalin as an aqueous solution, or between active pharmaceutical ingredients and their salt forms in drug products, introduce ambiguities that complicate data integration and reuse. RefChem, the PubChem Reference Collection, was introduced to address these challenges by disambiguating and semantically organizing chemical substances using standardized structures, names, and identifiers.

In this work, we present an RDF-based representation of RefChem to enable formal semantic integration within PubChemRDF and across external biomedical resources. We describe a modeling approach that captures core RefChem concepts including substance identity, structural relationships, naming types, associations and variants with established Semantic Web technologies. The approach leverages existing vocabularies where possible and introduces extensions to represent chemical relationships. We demonstrate how this RDF representation supports cross-domain linking, querying, and reasoning over chemical entities, enabling more precise integration with heterogeneous datasets such as drugs, toxicology data, biological assays, literature, patents, etc. This work highlights the potential of combining RefChem with RDF and ontology-based modeling to advance the interoperability and FAIRness of chemical data integration across a broad ecosystem of biomedical knowledge graphs.

Acknowledgements

This work was supported by the National Center for Biotechnology Information of the National Library of Medicine (NLM), National Institutes of Health (NIH). The contributions of the NIH author(s) are considered Works of the United States Government. The findings and conclusions presented in this paper are those of the author(s) and do not necessarily reflect the views of the NIH or the U.S. Department of Health and Human Services.